

Air Pollution ·

Air pollution is one of the most dangerous forms of environmental pollution in current times. As a result of natural and man-made activities, gases such as CO, SO₂, H₂S, and oxides of nitrogen, mist particulates, and aerosol.

❑ Causes of Air Pollution

- Industries
 - Automobile and Domestic Fuels
 - High Proportion of undesirable
- ❑ Gases, such as sulphur dioxide and carbon dioxide



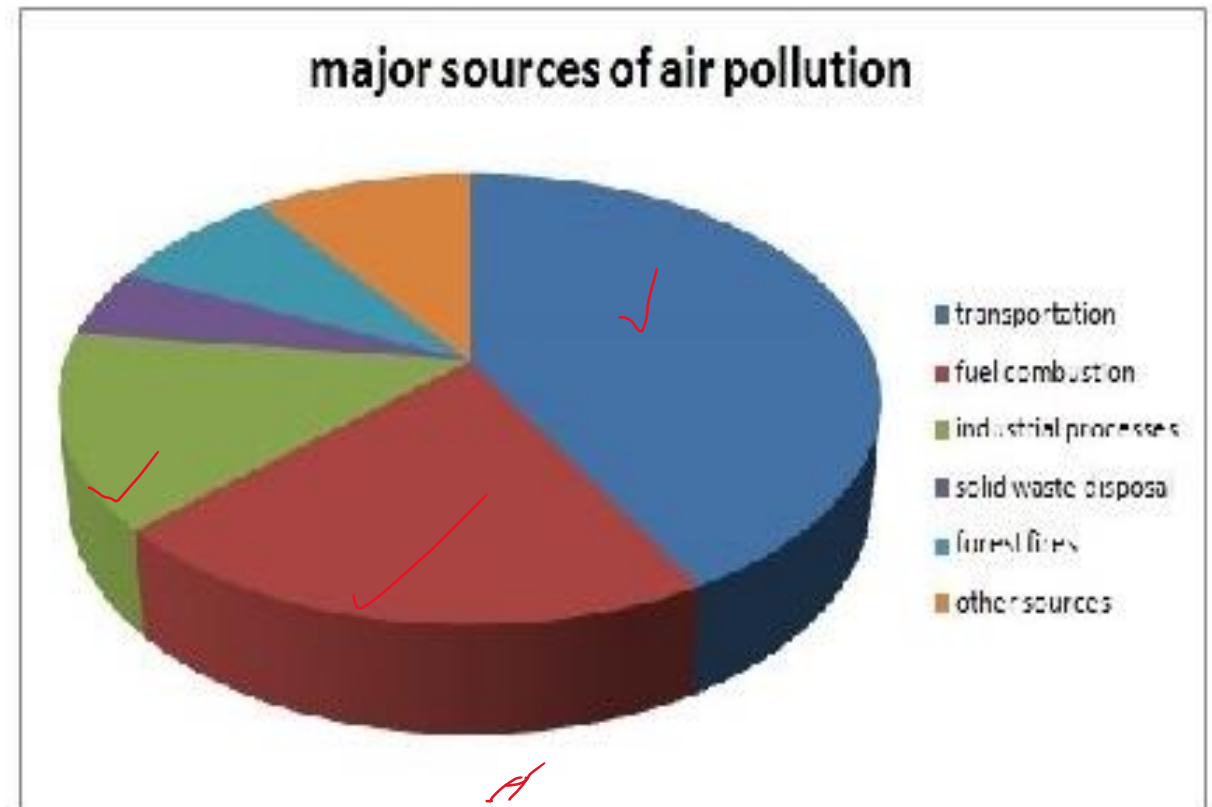
Sources of Air Pollution

Sources of air pollution can be classified into two category:

- Man-made
- Natural

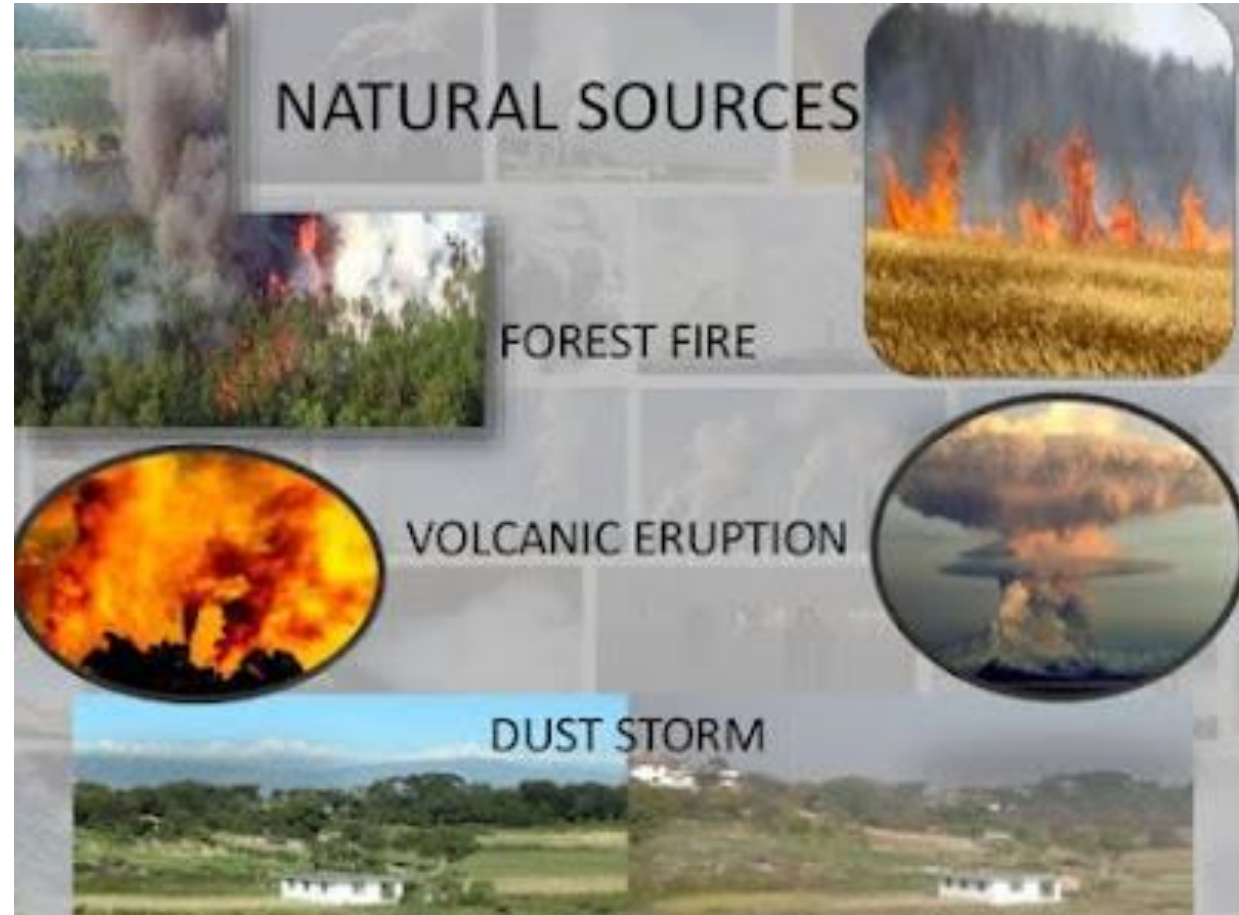
I. Man-Made Sources

- ❖ Population explosion
- ❖ Burning of fossil fuels
- ❖ Vehicular discharge
- ❖ Rapid Industrilization
- ❖ Agricultural activities
- ❖ Modern warfare
- ❖ Smoking



Natural sources

- Forest Fire
- Volcanic eruption
- Dust storm



Types of Air Pollutants

- **Primary Pollutants**

A primary pollutant can be defined as a harmful chemical that directly enters the air as a result of either natural or man-made activities.

- Oxides of carbon:** Carbon monoxide: It is a colourless, odourless, non-irritating but very poisonous gas. It is a product by incomplete combustion of fuel such as natural gas, coal or wood. Vehicular exhaust is a major source of carbon monoxide. iv. Carbon dioxide (CO₂): A greenhouse gas emitted from combustion but is also a gas vital to living organisms. It is natural gas in the atmosphere.
- Nitrogen oxides:** Especially nitrogen dioxide is emitted from high-temperature combustion. Nitrogen dioxide is the chemical compound with the formula NO₂. It is responsible for photochemical smog, acid rain etc
- Hydrocarbon:** VOCs are an important outdoor air pollutant. In this field they are often divided into the separate categories of methane (CH₄) and non-methane (NMVOCs). Methane is an extremely efficient greenhouse gas that contributes to enhanced global warming. Other hydrocarbon VOCs are also significant greenhouse gases via their role in creating ozone and in prolonging the life of methane in the atmosphere, although the effect varies depending on local air quality. Within the NMVOCs, the aromatic compounds benzene, toluene and xylene are suspected carcinogens and may lead to leukaemia through prolonged exposure. 1, 3-butadiene is another dangerous compound which is often associated with industrial uses.
- Sulphur oxides:** Sulphur oxides (SO_x): SO₂ is produced by volcanoes and in various industrial processes. Since coal and petroleum often contain sulphur compounds, their combustion generates sulphur dioxide. Further oxidation of SO₂, usually in the presence of a catalyst such as NO₂, forms H₂SO₄, and thus acid rain. This is one of the causes for concern over the environmental impact of the use of these fuels as power sources.
- Particulate Matter:** Particulates alternatively referred to as particulate matter (PM) or fine particles, are tiny particles of solid or liquid suspended in a gas. In contrast, aerosol refers to particles and the gas together. Sources of particulate matter can be manmade or natural. Some particulates occur naturally, originating from volcanoes, dust storms, forest and grassland fires, living vegetation, and sea spray. Human activities, such as the burning of fossil fuels in vehicles, power plants and various industrial processes also generate significant amounts of aerosols.

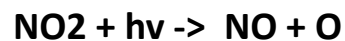
Secondary Pollutants

Secondary pollutants are formed by primary pollutants through their reaction with normal atmospheric compounds

- **Ozone (O_3):** Ozone is a secondary pollutant formed by photochemical reaction between primary pollutants and natural atmospheric gas. Ozone affects the respiratory and nervous system.
- **Sulfuric acid and nitric acid (component of acid rain):** Acid rain forms when water in the air combines with nitrogen oxides and sulfur dioxide (two types of pollutants) and then falls down the surface of the Earth. It has many damaging effects on vegetation, lakes, fish, buildings and other structures.
- **Smog:** When ultraviolet light from the sun reacts with nitrogen oxides in the atmosphere, photochemical smog is produced.
- **Particulate matter:** Particulate matter or simply PM is a combination of solid particles and liquid droplets that can be found in the air. PM contains hazardous elements such as arsenic, beryllium, cadmium, chromium, lead, manganese, and nickel. PM causes significant health problems in humans
- **Peroxyacyl nitrates (PANs):** It is produced in the atmosphere when oxidized volatile organic compounds combine with nitrogen oxide. It is a component of photochemical smog.

Nitrogen oxides (NO_x): NO, NO₂, N₂O, N₂O₃, N₂O₅

- Nitrogen dioxide is emitted from high temperature combustion, automobile exhausts, forest fires, electric generation plants, smelting plants, industrial boilers, petroleum refineries, chemical industries and volcanic eruptions
- Natural Sources : During lightening discharge N₂ and O₂ in air combines to form NO. NO₂ is formed by fixing of nitrogen from atmosphere as well as from organic matter present in earth crust .
- N₂ + O₂ → 2NO
- Man Made Sources :
- N₂ + O₂ ⇌ 2NO (1200-1700°C) Combustion of fossil fuel
- 2NO + O₂ → 2NO₂ (1100°C)
- NO + O₃ → NO₂ + O₂ (reaction with ground level ozone)
- Sinks for NO_x : NO and NO₂ undergoes various photochemical and chemical reaction in the atmosphere) lading to formation of HNO₃. Thus all the pollutants get precipitated in the form of nitrates during rainfall . Formation and destruction of NO₂ as well as O₃ is a cyclic process taking place in atmosphere .



- In the presence of VOC , NO_x reacts with water in presence of Oxygen resulting in the formation of HNO₃.



- 2NO + O₃ → N₂O₅ + O₂
- N₂O₅ + H₂O → 2HNO₃
- N₂O₅ reacts rapidly with liquid water in aerosol particles This nitric acid contributes to acid rain or may deposit to soil, where it makes nitrate, which is of use to growing plants.

Oxides of Nitrogen contd

Nitrogen dioxide is an irritant gas, NO_x mainly impacts on respiratory conditions causing inflammation of the airways at high levels. Long term exposure can decrease lung function, increase the risk of respiratory conditions and increases the response to allergens. NO_x also contributes to the formation of fine particles (PM) and ground level ozone, both of which are associated with adverse health effects.

NO_x gases react to form smog and acid rain as well as being central to the formation of fine particles (PM) and ground level ozone, both of which are associated with adverse health effects.

Control of NO_x :

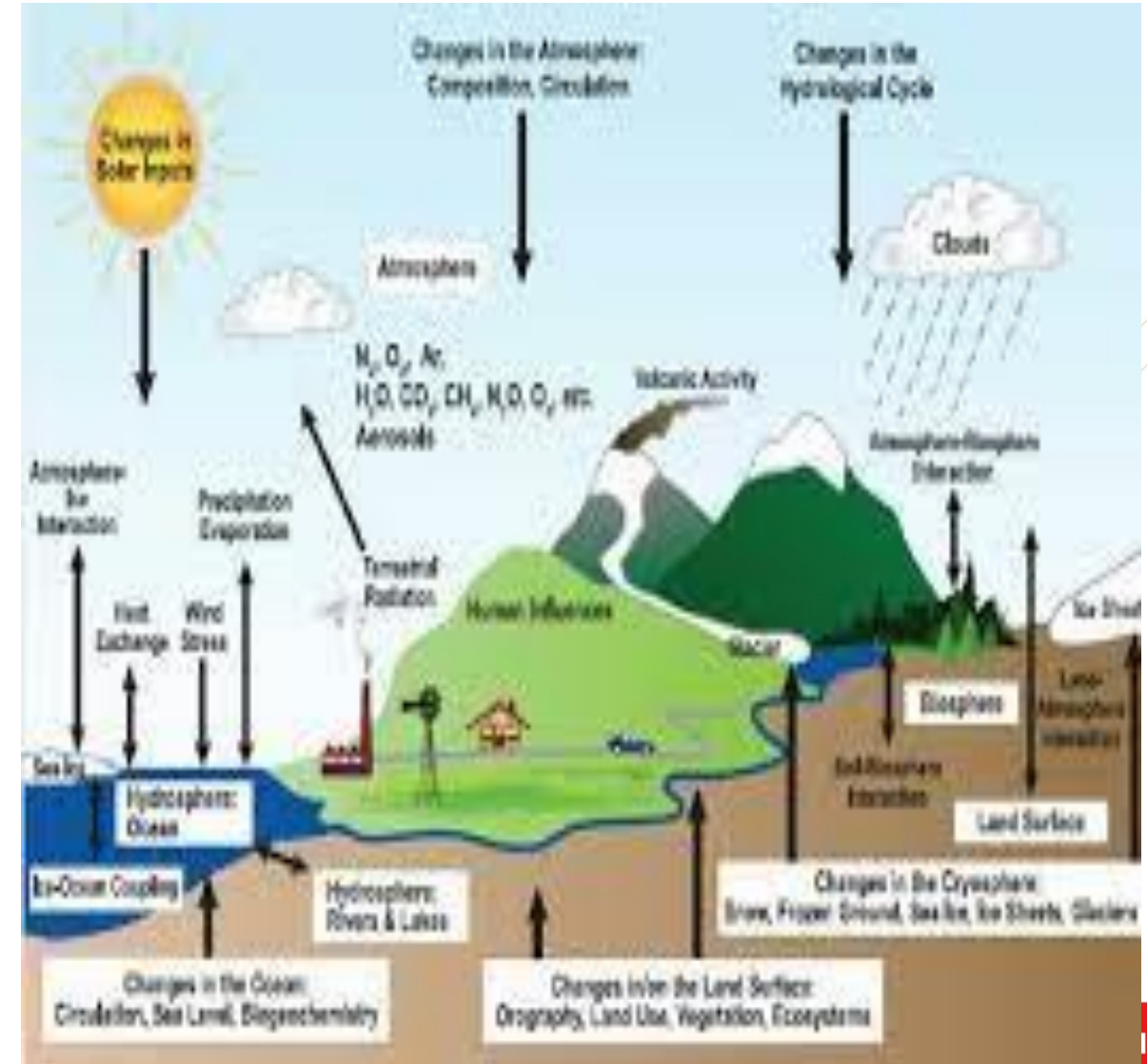
- Modification of Engine design : The burners are modified so that fuel air mixture burn slowly reducing temperature and intensity of combustion.
- Employing two stage combustion process (as discussed in CO)
- I stage NO_x → N₂ + NH₃ {catalyst Pt, Pd and Ru on ceramic material , CO gas
II stage CO → CO₂ (Pt and Ru on ceramic material)
- Scrubbing of flue gas with H₂SO₄

Carbon monoxide: (max permissible conc 40ppm)

- It is a colourless, odourless, non-irritating, most abundant but very poisonous gas. Source : It is a product by incomplete combustion of fuel such as natural gas, coal or wood. Vehicular exhaust is a major source of carbon monoxide. Photochemical reactions in the atmosphere, biological oxidation by marine organisms, etc.
- Sink: There is continued increase in CO emission but amount of CO in atmosphere remains relatively constant suggesting that a scavenging process or sink exist in atmosphere. Microorganism present in the soil converts significant amount of CO to CO₂
- Effects: Carbon monoxide affects the respiratory activity as haemoglobin has more affinity for CO than for oxygen (about 250 times). Thus, CO interferes with the blood's ability to carry oxygen to different part of body. CO combines with HB and thus reduces the oxygen-carrying capacity of blood.
 - $\text{Hb} + \text{O}_2 \rightarrow \text{HbO}_2 \rightarrow \text{Hb} + \text{O}_2$ (Hb is an oxygen carrier through
 - $\text{Hb} + \text{CO} \rightleftharpoons \text{HbCO}$ which different body cells intake O₂)
- This results in blurred vision, headache, unconsciousness and death due to asphyxiation (lack of oxygen).
- Asphyxia or suffocation may result if the oxygen concentration is reduced to below 19.5% by displacement
- Control of CO :
 1. Modification of Engine design: Engine design should be modified to maintain right proportion of oxygen in fuel air ratio for complete combustion of fuel.
 2. Fuel Modification: Fuel including natural gas, methane and blends of lighter hydrocarbon is recommended
 3. Treatment of exhaust gases : two stage catalytic converter preferred
 - I stage $\text{NO}_x \rightarrow \text{N}_2 + \text{NH}_3$ {catalyst Pt, Pd and Ru on ceramic material, CO gas
 - II stage $\text{CO} \rightarrow \text{CO}_2$ (Pt and Ru on ceramic material)

Air pollution and Biosphere

- Atmospheric pollutants are mainly present to a great extent in the **troposphere and the lower stratosphere**. Depending on the extent of air pollution the atmosphere can be divided into three layers.
- The first layer to 100m from the ground is highly polluted in general and in urban and industrial area in particular. Pollutants present in this layer absorbed by vegetation, water surface, and buildings.
- The next layer is 100-2000m high, contains lesser amounts of pollutants. Turbulent air currents, drizzle, rain, and fog divert and dilute the concentration of pollutants there.
- The third layer containing a large amount of water vapour and clouds, extends up to the troposphere. Here cloud water droplets dissolve some pollutants and bring them back to the earth as rain.
- The secondary pollutants formed in the atmosphere by the reaction of primary pollutants, water, oxygen, and UV rays of the sun in the atmosphere affect the soil, vegetation, crops, animals, and human beings.



Effect of Air pollution on Meteorology

- It can be defined as the study of behaviour of atmospheric variation that determines the extent of air pollution.
- Factors like wind, whose speed and direction of flow directly determine the extent of dispersion and dilution of air pollutants are called primary meteorology factor
- Precipitation, humidity, and solar radiation that control the dispersion of the pollutants indirectly are called secondary meteorological factors.

- **Wind speed and direction**

When high pollutant concentrations occur at a monitoring station, wind data records can determine the general direction and area of the emissions. Identifying the sources means planning to reduce the impacts on air quality.

An instrument called an anemometer measures wind speed. At monitoring stations, the type of anemometer used is a sonic anemometer.

A sonic anemometer operates on the principle that the speed of wind affects the time it takes for sound to travel from one point to another. Sound travelling with the wind will take less time than sound travelling into the wind. By measuring sound wave speeds in 2 different directions at the same time, sonic anemometers can measure both wind speed and direction.

- **Temperature**

Measuring temperature supports air quality assessment, air quality modeling, and forecasting activities. Temperature and sunlight (solar radiation) play an important role in the chemical reactions that occur in the atmosphere to form photochemical smog from other pollutants. Favourable conditions can lead to increased concentrations of smog. The most common way of measuring temperature is to use a material with a resistance that changes with temperature, such as platinum wire. A sensor measures this change and converts it into a temperature reading.

- **Humidity**

Like temperature and solar radiation, water vapour plays an important role in many thermal and photochemical reactions in the atmosphere. As water molecules are small and highly polar, they can bind strongly to many substances. If attached to particles suspended in the air they can significantly increase the amount of light scattered by the particles ([measuring visibility](#)). If the water molecules attach to corrosive gases, such as sulfur dioxide, the gas will dissolve in the water and form an acid solution that can damage health and property

Air Quality Index

The Central Pollution Control Board (CPCB) initiated its own National Ambient Air Quality Monitoring (NAAQM) program in 1985

Ambient air quality standards in India developed by the Central Pollution Control Board

Area Category	SPM $\mu\text{g}/\text{m}^3$	SO ₂ $\mu\text{g}/\text{m}^3$	Co $\mu\text{g}/\text{m}^3$	NO _x $\mu\text{g}/\text{m}^3$
Industrial and mixed use	500	120	5000	120
Residential and rural	200	80	2000	80
Sensitive	100	3	1000	30

Air Quality Control Technique

Following measures have been suggested to control air pollution-

- (i) Some gases, which are more soluble in a particular liquid than air, for example, ammonia in water, can be separated by dissolving in it
- (ii) Particles larger than 50 mm are separated in gravity settling tanks.
Using cyclone collectors or electrostatic precipitators separates fine particles.
- (iii) The height of chimneys should be increased to the highest possible level to reduce pollution at the ground level.
- (iv) SO₂ pollution can be controlled by extracting sulphur from the fuel before use.
- (v) Pollution control laws should be enforced strictly.
- (vi) Trees should be planted on the roadside, riverbanks, parks and open places as they keep the environment fresh.

Common air pollution control technologies:

- Cyclones.
- Scrubbers.
- Settling chambers
- Filters.
- Absorption.
- Electrostatic precipitation.
- Adsorption
- Combustion.